

**Board of Accreditation for Engineering and
Technical Education (BAETE)**

**ACCREDITATION CRITERIA
(Version 3.0)**

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ACC-TMP-04-04-V3.0

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ACCREDITATION CRITERIA

1.0 PURPOSE OF THIS DOCUMENT

The purpose of this document is to describe the nine accreditation criteria that are the basis for evaluating a program by the Board of Accreditation for Engineering and Technical Education (BAETE) of the Institution of Engineers, Bangladesh (IEB).

2.0 SCOPE OF THIS DOCUMENT

The accreditation criteria presented here apply to the Higher Educational Institutions (HEI) offering undergraduate engineering programs in Bangladesh.

3.0 ROLES AND RESPONSIBILITIES

3.1 Engineering programs applying for BAETE Accreditation

- Prepare a Self-assessment report (SAR) based on the ten accreditation criteria. The Self-assessment report template for the programs must be prepared as per ACC-TMP-04-04 (V2.1 – effective till 30 June 2026; V3.0 – effective from 1 July 2025; programs may apply using template V2.1 or V3.0 between 1 July 2025 and 30 June 2026).

3.2 BAETE

- Conducts the program evaluation based on the accreditation criteria presented in this document.

4.0 DEFINITIONS AND ACRONYMS

See ACC-MAN-06 for definitions and acronyms.

5.0 ACCREDITATION CRITERIA

This section presents the nine accreditation criteria, which are the basis for evaluating a program for accreditation. The following subsections sequentially outline the required attributes of each criterion

in terms of sub-criteria. This section does not provide any quantitative benchmark requirement to satisfy a criterion or sub-criterion adequately or appropriately. Adequacy or appropriateness is to be decided qualitatively, considering what is necessary for the students to attain the outcomes. It should be noted that no sub-criterion is assigned any weight. Each criterion must be holistically evaluated in terms of the qualitative benchmark requirements. The report template that the evaluation teams shall use is given in ACC-TMP-04-05 V2.1, V3.0.

5.1 Program Educational Objectives

Program Educational Objectives (PEOs) are broad statements describing the career and professional accomplishments the program prepares graduates to achieve. PEOs are assessable based on the attributes and accomplishments of graduates, preferably those who have worked for 3 to 5 years after graduation. Requirements under this criterion are described in terms of the following sub-criteria.

- i. Published PEOs must be clear, concise, assessable, and realistic within the context of the available resources.
- ii. PEOs must be consistent with the vision and mission of the institution or the department offering the program.
- iii. The program must demonstrate an established system for compiling the level of attainment of PEOs, including a mechanism for tracking and obtaining feedback from graduates and their employers.

5.2 Program Outcomes and Assessment

Program Outcomes (POs) or graduate attributes are narrower statements describing what students are expected to know and be able to do by graduation. These statements relate to the knowledge, skills, and attitudes students acquire while progressing through the program. By graduation, the program must demonstrate that students have achieved an acceptable minimum level of specific knowledge, skills, and behavioral traits. The BAETE requires that students acquire the following POs or graduate attributes:

Program Outcome	Description
PO1	Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions of complex engineering problems.
PO2	Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development (WK1 to WK4).
PO3	Design creative solutions for complex engineering problems and design systems, components or processes to meet identified needs with appropriate consideration for public health and safety, whole-life cost, net zero carbon as well as resource, cultural, societal, and environmental considerations as required (WK5).
PO4	Conduct investigations of complex engineering problems using research methods including research-based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions (WK8).
PO5	Create, select and apply and recognize limitations of appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering problems (WK2, WK6).
PO6	When solving complex engineering problems, analyze and evaluate sustainable development impacts to: society, the economy, sustainability, health and safety, legal frameworks, and the environment (WK1, WK5, and WK7).
PO7	Apply ethical principles and commit to professional ethics and norms of engineering practice and adhere to relevant national and international laws. Demonstrate an understanding of the need for diversity and inclusion (WK9).
PO8	Function effectively as an individual, and as a member or leader in diverse and inclusive teams and in multi-disciplinary, face-to-face, remote and distributed settings (WK9).
PO9	Communicate effectively and inclusively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend

	and write effective reports and design documentation, make effective presentations, taking into account cultural, language, and learning differences.
PO10	Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team and to manage projects and in multidisciplinary environments.
PO11	Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change (WK8).
PO12	Demonstrate knowledge and understanding of the competences necessary to transform opportunities and ideas into a new business.

In addition to incorporating the above-listed POs, the educational institution may include additional outcomes in its learning programs. An engineering program that aims to attain the POs mentioned above must ensure that its curriculum encompasses all the attributes of the Knowledge Profile (WK1 – WK9) presented in Table 6.1 (Section 6.0) and as included in the PO statements. The ranges of Complex Problem Solving (WP1 – WP7) and Complex Engineering Activities (EA1 – EA5) are given in Tables 6.2 and 6.3 (Section 6.0), respectively.

The required attributes under this criterion are described in the following sub-criteria.

- i. The process involved in defining the POs must be described. POs specified by the program must be significantly equivalent to the twelve graduate attributes or POs of BAETE.
- ii. Each course must have clear quality requirements and facilitate the achievement of course outcomes (COs) through teaching-learning and assessment methods.
- iii. The program must demonstrate the availability of the lecture plan, including COs, course content, books, grading policy, assessment tools and samples of corresponding student works, and assessment of CO attainment.
- iv. Students' academic performance must be continuously monitored to ensure the achievement of outcomes.

- v. The program must demonstrate, using direct methods, that the students attain all POs by graduation.

5.3 Curriculum and Teaching-Learning Processes

Requirements under this criterion are described in terms of the following sub-criteria.

- i. The curriculum must satisfy the relevant program-specific criteria.
- ii. The breadth and depth of the curriculum must be appropriate for solving complex engineering problems.
- iii. The curriculum must contain adequate courses on mathematics, natural sciences, computing, humanities, social sciences, and other non-engineering subjects.
- iv. Curriculum and teaching-learning processes must support the attainment of PEOs.
- v. The correlation between the course outcomes (COs) and POs must be appropriate.
- vi. How each attribute of the Knowledge Profile (WK1 – WK9) is addressed in the curriculum must be demonstrated through mapping. The program must also demonstrate how the attributes of the Range of Complex Engineering Problems (WP1 – WP7) and Complex Engineering Activities (EA1 – EA5) are incorporated into the teaching, learning, and assessment. Additionally, the program must demonstrate how various United Nations Sustainable Development Goals are considered in teaching, learning, and assessment.
- vii. The teaching-learning processes and assessment tools selected for each course must be effective and appropriate for achieving the relevant outcomes, including those pertaining to complex engineering problems and activities, if applicable.

5.4 Interactions with the Industry

A communication channel between the educational institution and the industry should exist. The industry should be encouraged to provide feedback concerning the quality of the teaching-learning process. Requirements under this criterion are described in terms of the following sub-criteria.

- i. The industry must participate in developing the curriculum to ensure that it is relevant, regularly updated, and meets the needs of the industry, particularly in areas experiencing rapid changes.

- ii. The program must provide students with the opportunity to obtain industrial experience.

5.5 Continuous Quality Improvement

Requirements under this criterion are described in terms of the following sub-criteria.

- i. The program must have a functional quality assurance system under an institutional framework.
- ii. The program must address the concerns, weaknesses, and deficiencies the BAETE evaluation team identified in the last review.
- iii. The students receive periodic feedback on their academic performance from the course teachers.
- iv. The findings of the CQI exercises for PEOs must be evaluated, and the identified shortcomings and limitations must be used regularly to refine and improve the program.
- v. POs must be assessed on a regular cycle. The program must undertake regular appropriate review of POs, considering feedback from relevant stakeholders.
- vi. The program must evaluate the curriculum and teaching quality regularly while considering feedback from faculty members, students and other stakeholders, and use the results of these evaluations for continuous improvements.

5.6 Students

Requirements under this criterion are described in terms of the following sub-criteria.

- i. The program must have and practice published policies for the admission and transfer of students into the program.
- ii. The admission requirements must be appropriate for selecting students with the potential to achieve the program outcomes, including good grades in mathematics and natural sciences.
- iii. Every student must be assigned an advisor. The advisor must counsel, guide, and mentor the student on all academic and professional matters.
- iv. Students' workload must enable them to participate in extra- and co-curricular activities and the activities of relevant professional societies.

5.7 Faculty

Requirements under this criterion are described in terms of the following sub-criteria.

- i. The department must have a sufficient number of full-time faculty members to ensure that they are not overloaded with courses and that the program does not become overly dependent on part-time faculty members.
- ii. The department must have a good blend of senior and junior faculty members with adequate academic qualifications and specializations in areas closely related to the program.
- iii. The teacher-student ratio, class size, and teaching load must not compromise opportunities for interactions between the students and faculty members.
- iv. Faculty members must be responsible for designing and updating the curriculum, establishing course and program outcomes, and selecting and using appropriate assessment tools for evaluating student performance in classes and the achievement of outcomes.
- v. Faculty members must be engaged in research, development and/or professional activities.
- vi. All the faculty members must be adequately trained to establish course outcomes, conduct teaching-learning activities that are appropriate for the outcomes and assess the level of outcome achievement.

5.8 Governance, Finance and Safety

Requirements under this criterion are described in terms of the following sub-criteria.

- i. The officers and statutory committees of the institution must function effectively as per their roles defined in the institutional act/statute.
- ii. The administrative and academic policies must be put into practice.
- iii. The financial resources committed to the program must be adequate for the appropriate functioning of the program, including recruiting and retaining qualified faculty members and procuring the necessary lab equipment and tools to support teaching and learning.
- iv. A safety plan must exist and be practiced to address the risk from manmade or natural hazards, including fire detection and suppression, as well as incidents and accidents in the laboratories.

5.9 Academic Facilities and Technical Support

Requirements under this criterion are described in terms of the following sub-criteria.

- i. The institution must have a well-stocked library with adequate books, e-books, journals, and other relevant resources for the program.
- ii. The learning environment must be adequate and conducive.
- iii. Every student must have the opportunity to work in the laboratories.
- iv. Students and faculty members must have access to adequate computing and Internet facilities, including hardware, software tools, and support.

6.0 TABLES

Table 6.1: Knowledge and attitude Profile

	Attribute
WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences
WK2	Conceptually based mathematics, numerical analysis, data analysis, statistics and the formal aspects of computer and information science to support detailed analysis and modeling applicable to the discipline
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline
WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as professional responsibility of an engineer to public safety and sustainable development
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues
WK9	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes

Table 6.2: Range of Complex Engineering Problem Solving

Attribute	Complex Engineering Problems have characteristics WP1 and some or all of WP2 to WP7:
Depth of knowledge required	WP1: Cannot be resolved without in-depth engineering knowledge at the level of one or more of WK3, WK4, WK5, WK6 or WK8 which allows a fundamentals-based, first principles analytical approach
Range of conflicting requirements	WP2: Involve wide-ranging or conflicting technical, non-technical issues (such as ethical, sustainability, legal, political, economic, societal) and consideration of future requirements
Depth of analysis required	WP3: Have no obvious solution and require abstract thinking, creativity and originality in analysis to formulate suitable models
Familiarity of issues	WP4: Involve infrequently encountered issues or novel problems
Extent of applicable codes	WP5: Address problems not encompassed by standards and codes of practice for professional engineering
Extent of stakeholder involvement and conflicting requirements	WP6: Involve collaboration across engineering disciplines, other fields, and/or diverse groups of stakeholders with widely varying needs
Interdependence	WP7: Address high level problems including many components or sub-problems that may require a systems approach

Table 6.3: Range of Complex Engineering Activities

Attribute	Complex activities mean (engineering) activities or projects that have some or all of the following characteristics:
Range of resources	EA1: Involve the use of diverse resources including people, data and information, natural, financial and physical resources and appropriate technologies including analytical and/or design software
Level of interactions	EA2: Require optimal resolution of interactions between wide-ranging and/or conflicting technical, non-technical, and engineering issues
Innovation	EA3: Involve creative use of engineering principles, innovative solutions for a conscious purpose, and research-based knowledge
Consequences to society and the environment	EA4: Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation
Familiarity	EA5: Can extend beyond previous experiences by applying principles-based approaches

7.0 The BAETE Accreditation Manual, Standard Operating Procedures, Templates, Flowcharts, Schedule, and Work Instructions

See [ACC-MAN-00](#) for details.

8.0 REVISION HISTORY

Date	Version	Description of Change
23 March 2017	1.0	Initial Release
05 March 2019	2.0	Year-long Final Year Design Project or Capstone Project is no longer obligatory in the curriculum. Demonstration of culminated POs in solving complex engineering problems remains required, and the year-long Final Year Design Project or Capstone Project is the preferred method. The descriptions of POs have been updated following IEA Version 3.21, June 2013. Requirements of the Knowledge Profiles (K1-K8) have been explicitly mentioned in the description of each PO. The tables describing the Range of Complex Engineering Problem-Solving and Range of Complex Engineering Activities have been updated following IEA Version 3.21, June 2013. The program should map out how each attribute of the Knowledge Profile (K1 – K8) is addressed in the curriculum. The program should also demonstrate how each attribute of the Range of Complex Engineering Problems (P1 – P7) and Complex Engineering Activities (A1 – A5) is incorporated in teaching, learning and assessment. Results of evaluation of PO attainment should be reported.
16 May 2022	2.1	The required and desired attributes of each of the ten criteria have been listed in terms of sub-criteria. The required attributes are indicated through the word must, while the desired attributes are indicated through the word should. Section 4.8: Titles for the POs have been removed. Reference to Annex III Evaluation Team Report (ETR) template has been added in the first paragraph of chapter 4.
23 April 2024	2.2	The purpose and scope of the document were added; roles and responsibilities were added; definitions and acronyms were added; there were no changes in the previous content.
23 April 2024	3.0	Released as a standalone document; desired attributes are removed from the criteria; criteria are rearranged, and criteria relating to outcomes are brought forward; Organization and Governance (criterion 1 in v2.1) and Financial and Physical Resources (criterion 2 in v2.1) were merged as “Governance, Finance and Safety”; sub-criteria under each criterion were reviewed and revised.

Appendix A: The BAETE Accreditation Manual Updates

Version 2.2 (July 2024) to Version 3.0 (July 2025)

Knowledge Profiles changed from **K** to **WK**, Complex Problem Solving from **P** to **WP**, and Complex Activities from **A** to **EA**.

Program Outcomes (POs) and Assessment

Attribute / No.	Version 2.2	Version 3.0
PO1: Engineering Knowledge	Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization as specified in K1 to K4 respectively to the solution of complex engineering problems.	Apply knowledge of mathematics, natural science, computing , engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions of complex engineering problems.
PO2: Problem Analysis	Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences (K1 to K4).	Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development (WK1 to WK4) .
PO3: Design/ Development of Solutions	Design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations (K5).	Design creative solutions for complex engineering problems and design systems, components or processes to meet identified needs with appropriate consideration for public health and safety, whole-life cost, net zero carbon as well as resource , cultural, societal, and environmental considerations as required (WK5) .

Attribute / No.	Version 2.2	Version 3.0
PO4: Investigation	Conduct investigations of complex problems using research-based knowledge (K8) and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.	Conduct investigations of complex engineering problems using research methods including research-based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions (WK8).
PO5: Modern Tool Usage	Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering problems, with an understanding of the limitations (K6).	Create, select and apply <i>and recognize limitations of</i> appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering problems (WK2, WK6).
PO6: The Engineer and Society	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems (K7).	<i>When solving complex engineering problems, analyze and evaluate sustainable development impacts to: society, the economy, sustainability, health and safety, legal frameworks, and the environment (WK1, WK5, and WK7).</i>
PO7: Environment and Sustainability / Ethics (V2.2 PO7 focused on Sustainability; V3 PO7 shifts to Ethics).	Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts (K7).	Apply ethical principles and commit to professional ethics and <i>norms of engineering practice and adhere to relevant national and international laws. Demonstrate an understanding of the need for diversity and inclusion (WK9).</i>
PO8: Ethics / Individual and Teamwork	Apply ethical principles and commit to professional ethics and responsibilities and	Function effectively as an individual, and as a member or leader in <i>diverse and inclusive</i> teams and in multi-

Attribute / No.	Version 2.2	Version 3.0
(V2.2 PO8 focused on Ethics, V3 PO8 shifts to Individual and Teamwork).	norms of engineering practice (K7).	disciplinary, <i>face-to-face, remote and distributed settings (WK9).</i>
PO9: Individual and Teamwork / Communication (V2.2 PO9 focused on Teamwork, V3 PO9 shifts to Communication).	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.	Communicate effectively <i>and inclusively</i> on complex engineering activities with the engineering community and with society at large... <i>taking into account cultural, language, and learning differences.</i>
PO10: Communication / Management & Finance (V2.2 PO10 focused on Communication, V3 PO10 shifts to Communication Management & Finance).	Communicate effectively on complex engineering activities... make effective presentations, and give and receive clear instructions.	Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team <i>and to manage projects and in multidisciplinary environments.</i>
PO11: Management & Finance / Lifelong Learning (V2.2 PO11 focused on Management, V3 PO11 shifts to Lifelong Learning).	Demonstrate knowledge and understanding of engineering management principles and economic decision-making...	Recognize the need for, and have the preparation and ability for <i>i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking</i> in the broadest context of technological change (WK8).
PO12: Lifelong Learning / ideas, opportunities and business (V2.2 PO12 focused on Lifelong Learning, V3 PO12 shifts to ideas, opportunities and business).	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.	<i>Demonstrate knowledge and understanding of the competences necessary to transform opportunities and ideas into a new business.</i>

Knowledge and attitude Profile

	Version 2.2 (K)	Version 3.0 (WK)
WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline.	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences
WK2	Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modeling applicable to the discipline.	Conceptually based mathematics, numerical analysis, data analysis, statistics and the formal aspects of computer and information science to support detailed analysis and modeling applicable to the discipline
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline
WK5	Knowledge that supports engineering design in a practice area.	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline
WK7	Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as professional responsibility of an

	of engineering activity; economic, social, cultural, environmental and sustainability	engineer to public safety and sustainable development
WK8	Engagement with selected knowledge in the research literature of the discipline	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues
WK9		Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes

Range of Complex Engineering Problem Solving

Attribute	Version 2.2 (P1–P7)	Version 3.0 (WP1–WP7)
Depth of Knowledge	P1: Cannot be resolved without in-depth engineering knowledge at the level of K3, K4, K5, K6 or K8.	WP1: Cannot be resolved without in-depth engineering knowledge at the level of WK3, WK4, WK5, WK6 or WK8...
Conflicting Requirements	P2: Involve wide-ranging or conflicting technical, engineering and other issues.	WP2: Involve wide-ranging or conflicting technical, <i>non-technical issues (such as ethical, sustainability, legal, political, economic, societal) and consideration of future requirements.</i>
Depth of Analysis	P3: Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models.	WP3: Have no obvious solution and require abstract thinking, <i>creativity</i> and originality in analysis to formulate suitable models.
Familiarity of Issues	P4: Involve infrequently encountered issues.	WP4: Involve infrequently encountered issues <i>or novel problems.</i>
Applicable Codes	P5: Are outside problems encompassed by standards and codes of practice.	WP5: Address problems <i>not encompassed</i> by standards and codes of practice.
Stakeholder Involvement	P6: Involve diverse groups of stakeholders with widely varying needs.	WP6: Involve <i>collaboration across engineering disciplines, other fields, and/or</i> diverse groups of stakeholders with widely varying needs.
Interdependence	P7: Are high-level problems including many component parts or sub-problems.	WP7: Address high level problems including many components or sub-problems <i>that may require a systems approach.</i>

Range of Complex Engineering Activities

Attribute	Version 2.2 (A1–A5)	Version 3.0 (EA1–EA5)
Range of Resources	A1: Involve the use of diverse resources (people, money, equipment, materials, information and technologies).	EA1: Involve the use of diverse resources including people, <i>data and information, natural, financial and physical resources and appropriate technologies including analytical and/or design software.</i>
Level of Interactions	A2: Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues.	EA2: Require <i>optimal resolution</i> of interactions between wide-ranging and/or conflicting technical, <i>non-technical</i> , and engineering issues.
Innovation	A3: Involve creative use of engineering principles and research-based knowledge in novel ways.	EA3: Involve creative use of engineering principles, <i>innovative solutions for a conscious purpose</i> , and research-based knowledge.
Consequences	A4: Have significant consequences in a range of contexts... difficulty of prediction and mitigation.	EA4: Have significant consequences in a range of contexts... difficulty of prediction and mitigation.
Familiarity	A5: Can extend beyond previous experiences by applying principles-based approaches.	EA5: Can extend beyond previous experiences by applying principles-based approaches.

Appendix B: Bloom's Taxonomy

Bloom's Taxonomy was created in 1956 under the leadership of educational psychologist Dr Benjamin Bloom in order to promote higher forms of thinking in education, such as analyzing and evaluating concepts, processes, procedures, and principles rather than just remembering facts (rote learning). It is most often used when designing educational, training, and learning processes. Bloom identified three *domains* of educational activities or learning (Bloom, et al. 1956):

- **Cognitive:** mental skills (*knowledge*)
- **Affective:** growth in feelings or emotional areas (*attitude or self*)
- **Psychomotor:** manual or physical skills (*skills*)

Cognitive Domain:

The cognitive domain involves knowledge and the development of intellectual skills (Bloom, et al, 1956). This includes the recall or recognition of specific facts, procedural patterns, and concepts that serve in the development of intellectual abilities and skills. There are six major categories of cognitive processes, which are listed in order below, starting from the simplest to the most complex.

Category	Examples, key words (verbs), and technologies for learning (activities)
C 1: Remembering: Recall or retrieve previous learned information.	Examples: Recite a policy. Quote prices from memory to a customer. Recite the safety rules. Key Words: defines, describes, identifies, knows, labels, lists, matches, names, outlines, recalls, recognizes, reproduces, selects, states Technologies: book marking, flash cards, rote learning based on repetition, reading
C 2: Understanding: Comprehending the meaning, translation, interpolation, and interpretation of instructions and problems. State a problem in one's own words.	Examples: Rewrite the principles of test writing. Explain in one's own words the steps for performing a complex task. Translate an equation into a computer spreadsheet. Key Words: comprehends, converts, defends, distinguishes, estimates, explains, extends, generalizes, gives an example, infers, interprets, paraphrases, predicts, rewrites, summarizes, translates Technologies: create an analogy, participating in cooperative learning, taking notes, storytelling, Internet Search.

C 3: Applying: Use a concept in a new situation or unprompted use of an abstraction. Applies what was learned in the classroom into novel situations in the work place.	Examples: Use a manual to calculate an employee's vacation time. Apply laws of statistics to evaluate the reliability of a written test. Key Words: applies, changes, computes, constructs, demonstrates, discovers, manipulates, modifies, operates, predicts, prepares, produces, relates, shows, solves, uses Technologies: collaborative learning, create a process, blog, practice
C 4: Analyzing: Separates material or concepts into component parts so that its organizational structure may be understood. Distinguishes between facts and inferences.	Examples: Troubleshoot a piece of equipment by using logical deduction. Recognize logical fallacies in reasoning. Gathers information from a department and selects the required tasks for training. Key Words: analyzes, breaks down, compares, contrasts, diagrams, deconstructs, differentiates, discriminates, distinguishes, identifies, illustrates, infers, outlines, relates, selects, separates Technologies: Fishbowls, debating, questioning what happened, run a test
C 5: Evaluating: Make judgments about the value of ideas or materials.	Examples: Select the most effective solution. Hire the most qualified candidate. Explain and justify a new budget. Key Words: appraises, compares, concludes, contrasts, criticizes, critiques, defends, describes, discriminates, evaluates, explains, interprets, justifies, relates, summarizes, supports Technologies: survey, blogging
C 6: Creating: Builds a structure or pattern from diverse elements. Put parts together to form a whole, with emphasis on creating a new meaning or structure.	Examples: Write a company operations or process manual. Design a machine to perform a specific task. Integrates training from several sources to solve a problem. Revises and process to improve the outcome. Key Words: categorizes, combines, compiles, composes,

	creates, devises, designs, explains, generates, modifies, organizes, plans, rearranges, reconstructs, relates, reorganizes, revises, rewrites, summarizes, tells, writes Technologies: Create a new model, write an essay, network with others
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Affective Domain:

The affective domain (Krathwohl, Bloom, Masia, 1973) includes the manner in which we deal with things **emotionally, such as feelings, values, appreciation, enthusiasms, motivations, and attitudes.** The five major categories are listed from the simplest behavior to the most complex.

Category	Example and Key Words (verbs)
A 1: Receiving Phenomena: Awareness, willingness to hear, selected attention.	Examples: Listen to others with respect. Listen for and remember the name of newly introduced people. Key Words: acknowledge, asks, attentive, courteous, dutiful, follows, gives, listens, understands
A 2: Responds to Phenomena: Active participation on the part of the learners. Attend and react to a particular phenomenon. Learning outcomes may emphasize compliance in responding, willingness to respond, or satisfaction in responding (motivation).	Examples: Participates in class discussions. Gives a presentation. Questions new ideals, concepts, models, etc. in order to fully understand them. Know the safety rules and practice them. Key Words: answers, assists, aids, complies, conforms, discusses, greets, helps, labels, performs, presents, tells

A 3: Valuing: The worth or value a person attaches to a particular object, phenomenon, or behavior. This ranges from simple acceptance to the more complex state of commitment. Valuing is based on the internalization of a set of specified values, while clues to these values are expressed in the learner's overt behavior and are often identifiable.	Examples: Demonstrates belief in the democratic process. Is sensitive towards individual and cultural differences (value diversity). Shows the ability to solve problems. Proposes a plan to social improvement and follows through with commitment. Informs management on matters that one feels strongly about. Key Words: appreciates, cherish, treasure, demonstrates, initiates, invites, joins, justifies, proposes, respect, shares
A 4: Organization: Organizes values into priorities by contrasting different values, resolving conflicts between them, and creating an unique value system. The emphasis is on comparing, relating, and synthesizing values.	Examples: Recognizes the need for balance between freedom and responsible behavior. Explains the role of systematic planning in solving problems. Accepts professional ethical standards. Creates a life plan in harmony with abilities, interests, and beliefs. Prioritizes time effectively to meet the needs of the organization, family, and self. Key Words: compares, relates, synthesizes
A 5: Internalizes Values (characterization): Has a value system that controls their behavior. The behavior is pervasive, consistent, predictable, and most important characteristic of the learner. Instructional objectives are concerned with the student's general patterns of adjustment (personal, social, emotional).	Examples: Shows self-reliance when working independently. Cooperates in group activities (displays teamwork). Uses an objective approach in problem solving. Displays a professional commitment to ethical practice on a daily basis. Revises judgments and changes behavior in light of new evidence. Values people for what they are, not how they look. Key Words: acts, discriminates, displays, influences, modifies, performs, qualifies, questions, revises, serves, solves, verifies

Psychomotor Domain:

The psychomotor domain (Simpson, 1972) includes physical movement, coordination, and use of the motor-skill areas. Development of these skills requires practice and is measured in terms of speed, precision, distance, procedures, or techniques in execution. Thus, psychomotor skills range from manual tasks, such as digging a ditch or washing a car, to more complex tasks, such as operating a complex piece of machinery or dancing.

The seven major categories are listed from the simplest behavior to the most complex:

Category	Example and Key Words (verbs)
P 1: Perception (awareness): The ability to use sensory cues to guide motor activity. This ranges from sensory stimulation, through cue selection, to translation.	Examples: Detects non-verbal communication cues. Estimate where a ball will land after it is thrown and then moving to the correct location to catch the ball. Adjusts heat of stove to correct temperature by smell and taste of food. Adjusts the height of the forks on a forklift by comparing where the forks are in relation to the pallet. Key Words: chooses, describes, detects, differentiates, distinguishes, identifies, isolates, relates, selects.
P 2: Set: Readiness to act. It includes mental, physical, and emotional sets. These three sets are dispositions that predetermine a person's response to different situations (sometimes called mindsets).	Examples: Knows and acts upon a sequence of steps in a manufacturing process. Recognize one's abilities and limitations. Shows desire to learn a new process (motivation). NOTE: This subdivision of Psychomotor is closely related with the "Responding to phenomenal subdivision of the Affective domain." Key Words: begins, displays, explains, moves, proceeds, reacts, shows, states, volunteers.
P 3: Guided Response: The early stages in learning a complex skill that includes imitation and trial and error. Adequacy of performance is achieved by practicing.	Examples: Performs a mathematical equation as demonstrated. Follows instructions to build a model. Responds hand-signals of instructor while learning to operate a forklift. Key Words: copies, traces, follows, react, reproduce, responds

P 4: Mechanism (basic proficiency): This is the intermediate stage in learning a complex skill. Learned responses have become habitual and the movements can be performed with some confidence and proficiency.	Examples: Use a personal computer. Repair a leaking faucet. Drive a car. Key Words: assembles, calibrates, constructs, dismantles, displays, fastens, fixes, grinds, heats, manipulates, measures, mends, mixes, organizes, sketches.
P 5: Complex Overt Response (Expert): The skillful performance of motor acts that involve complex movement patterns. Proficiency is indicated by a quick, accurate, and highly coordinated performance, requiring a minimum of energy. This category includes performing without hesitation, and automatic performance. For example, players are often utter sounds of satisfaction or expletives as soon as they hit a tennis ball or throw a football, because they can tell by the feel of the act what the result will produce.	Examples: Maneuvers a car into a tight parallel parking spot. Operates a computer quickly and accurately. Displays competence while playing the piano. Key Words: assembles, builds, calibrates, constructs, dismantles, displays, fastens, fixes, grinds, heats, manipulates, measures, mends, mixes, and organizes, sketches. NOTE: The Key Words are the same as Mechanism, but will have adverbs or adjectives that indicate that the performance is quicker, better, more accurate, etc.
P 6: Adaptation: Skills are well developed and the individual can modify movement patterns to fit special requirements.	Examples: Responds effectively to unexpected experiences. Modifies instruction to meet the needs of the learners. Perform a task with a machine that it was not originally intended to do (machine is not damaged and there is no danger in performing the new task). Key Words: adapts, alters, changes, rearranges, reorganizes, revises, varies.
P 7: Origination: Creating new movement patterns to fit a particular situation or specific problem. Learning outcomes emphasize creativity based upon highly developed skills.	Examples: Constructs a new theory. Develops a new and comprehensive training programming. Creates a new gymnastic routine. Key Words: arranges, builds, combines, composes, constructs, creates, designs, initiate, makes, originates.

REVISED Bloom's Taxonomy Action Verbs

Definitions	I. Remembering	II. Understanding	III. Applying	IV. Analyzing	V. Evaluating	VI. Creating
Bloom's Definition	Exhibit memory of previously learned material by recalling facts, terms, basic concepts, and answers.	Demonstrate understanding of facts and ideas by organizing, comparing, translating, interpreting, giving descriptions, and stating main ideas.	Solve problems to new situations by applying acquired knowledge, facts, techniques and rules in a different way.	Examine and break information into parts by identifying motives or causes. Make inferences and find evidence to support generalizations.	Present and defend opinions by making judgments about information, validity of ideas, or quality of work based on a set of criteria.	Compile information together in a different way by combining elements in a new pattern or proposing alternative solutions.
Verbs	- Choose - Define - Find - How - Label - List - Match - Name - Omit - Recall - Relate - Select - Show - Spell - Tell - What - When - Where - Which - Who - Why	- Classify - Compare - Contrast - Demonstrate - Explain - Extend - Illustrate - Infer - Interpret - Outline - Relate - Rephrase - Show - Summarize - Translate	- Apply - Build - Choose - Construct - Develop - Experiment with - Identify - Interview - Make use of - Model - Organize - Plan - Select - Solve - Utilize	- Analyze - Assume - Categorize - Classify - Compare - Conclusion - Contrast - Discover - Dissect - Distinguish - Divide - Examine - Function - Inference - Inspect - List - Motive - Relationships - Simplify - Survey - Take part in - Test for - Theme	- Agree - Appraise - Assess - Award - Choose - Compare - Conclude - Criteria - Criticize - Decide - Deduct - Defend - Determine - Disprove - Estimate - Evaluate - Explain - Importance - Influence - Interpret - Judge - Justify - Mark - Measure - Opinion - Perceive - Prioritize - Prove - Rate - Recommend - Rule on - Select - Support - Value	- Adapt - Build - Change - Choose - Combine - Compile - Compose - Construct - Create - Delete - Design - Develop - Discuss - Elaborate - Estimate - Formulate - Happen - Imagine - Improve - Invent - Make up - Maximize - Minimize - Modify - Original - Originate - Plan - Predict - Propose - Solution - Solve - Suppose - Test - Theory

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